

Paper Practice 1.5**Absolute Value Expressions**

Show all of your work in the space provided. Simplify INSIDE the absolute value bars first. Circle your final answers.

- 1.** Let $a = 5$, $b = -9$, and $c = 3$. Evaluate each expression. (Absolute Value)

a. $|b|$

b. $-|a|$

c. $|-c|$

d. $-|-b|$

- 2.** Evaluate each expression. (Evaluate)

a. $|3x + 7|$, if $x = -4$

b. $|-5x - 3y|$, if $x = 2$ and $y = -4$

- 3.** Evaluate $|-3x + 4y|$, if $x = 6$ and $y = -2$.
(Evaluate)

- 4.** Evaluate $18 - |2 + 3x|$, if $x = 4$. (Evaluate)

- 5.** Evaluate $3|4x - 14| + 5$, if $x = 2$. (Evaluate)

- 6.** Evaluate $2.5 - 1.5|4y + 1.2|$, if $y = -2$. (Evaluate)

- 7.** The accuracy of a kitchen scale is the positive difference between the reading r and the true weight w . Write TWO expressions that both represent the accuracy. (Write)
- 8.** A scale reads 4.92 pounds for a bag labeled 5 pounds. Use your expression from #7 to find the accuracy. (Apply)
- 9.** Explain in one sentence why $|x|$ can never be negative, no matter what x is. (Explain)
- 10.** Error Analysis – Gus evaluated $-|-7|$ and wrote 7. Identify his mistake, then evaluate correctly. (Error Analysis)
- 11.** Peak A is taller than peak B. Which expression is NOT equivalent to the difference in their heights? Circle one and explain. (Reason)
- Ⓐ $|a - b|$ Ⓑ $|b - a|$ Ⓒ $|b| - |a|$
Ⓓ $|a| - |b|$
- 12.** A thermometer reads 98.5°F when the true temperature is 98.6°F . Write and evaluate an absolute value expression for the accuracy. (Apply)